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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	Adaptive AI-Based Interview Coaching System	
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Abstract

The Adaptive AI-Based Interview Coaching System is proposed as an intelligent software engineering solution for providing personalized, continuous, and adaptive interview preparation. The system is designed around the principles of Adaptive Software Engineering (ASE), in which software requirements, behavior, and learning strategies can evolve in response to changing user needs and feedback. Instead of providing the same fixed set of interview questions to every learner, the proposed system continuously observes the user's performance and adapts subsequent interview sessions accordingly. This approach makes the coaching process more flexible, user-centered, and responsive to individual learning progress.

The primary objective of the system is to provide an AI-supported environment where students and job seekers can practice technical, behavioral, and human-resources interview questions in a realistic manner. At the beginning of a session, the user can specify a target job role, domain, experience level, and preferred interview type. Based on these inputs, the system generates or selects suitable questions. During the interview, the user's responses are analyzed using Artificial Intelligence and Natural Language Processing techniques. The system evaluates aspects such as relevance of the answer, clarity, completeness, communication quality, confidence indicators, and technical correctness where applicable.

A key feature of the proposed system is its adaptive feedback mechanism. After each response, the system identifies strengths and weaknesses and uses this information to influence the next stage of the interview. For example, if a user performs well on basic questions, the system can progressively introduce more challenging questions. If the user repeatedly struggles with a particular topic, the system can provide targeted feedback and generate additional practice questions for that area. Thus, the software does not follow a completely static workflow; its behavior changes according to observed performance, making the coaching process adaptive.

The system follows an iterative software engineering approach in which user feedback, performance data, and changing requirements can be incorporated into subsequent development cycles. The architecture can be organized into modules such as user management, interview configuration, question generation, response processing, performance evaluation, adaptive difficulty management, feedback generation, and progress tracking. These modules allow the system to be modified and extended as new interview domains, question types, evaluation criteria, or AI capabilities are introduced. Such modularity supports adaptability at both the product level and the development level.

From an Adaptive Software Engineering perspective, the project emphasizes continuous learning and change management. Interview preparation requirements are not identical for all users and may change as the user improves or as industry expectations evolve. Therefore, the proposed system is intended to support

incremental refinement rather than depending on a single fixed set of requirements. Performance information collected during earlier sessions can be used to personalize later sessions, while feedback from users can guide improvements to the system's question bank, evaluation rules, and coaching strategies.

The proposed system can maintain a performance history containing scores, frequently missed concepts, communication-related observations, question difficulty, and progress across sessions. This information can be used to generate personalized recommendations such as topics to revise, interview types to practice, and difficulty levels to attempt next. A dashboard can present progress in a simple form so that users can understand their improvement over time. The system can therefore act not only as a mock interviewer but also as an adaptive learning assistant that supports continuous skill development.

The project also addresses limitations of conventional interview preparation, where practice may be repetitive, feedback may be delayed, and personalized human coaching may not always be available. By combining AI-based analysis with adaptive software engineering principles, the proposed solution aims to make interview practice more accessible and responsive. The system can provide immediate feedback after a session and can adjust future practice according to the learner's needs. This reduces dependence on a fixed training sequence and encourages users to repeatedly practice areas in which improvement is required.

Overall, the Adaptive AI-Based Interview Coaching System demonstrates the application of Adaptive Software Engineering to an AI-enabled educational and career-support platform. Its central idea is to build a system that can respond to uncertainty, changing user requirements, and evolving performance rather than remaining static after deployment. By integrating adaptive question selection, AI-based response evaluation, personalized feedback, performance tracking, and iterative improvement, the proposed system aims to improve interview readiness, communication skills, technical preparation, and confidence. The project therefore provides a practical example of how adaptive software development principles can be combined with Artificial Intelligence to create a flexible and user-centered coaching application.

Signature of the Guide HOD